DESIGN CODE OF FULL ADDER (VERILOG)

/home/dhruv/cao_lab/vivado_proj/cao_lab_u24ev057/rtl/rca_board/fa_verilog.v [Verilog]

```
1  `timescale 1ns / 1ps
2  //////////////////////////////////////////////////////////////////////////////////
   //
3  // Company:
4  // Engineer:
5  //
6  // Create Date: 10/01/2026 04:43:21 PM
7  // Design Name:
8  // Module Name: fa_verilog
9  // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 //////////////////////////////////////////////////////////////////////////////////
   //
21
22
23 module fa_verilog(
24
25     input a,
26     input b,
27     input cin,
28     output s,
29     output cout
30
31     );
32
33         assign s = a^b^cin;
34
35         assign cout = (a&b)|(b&cin)|(cin&a);
36
37 endmodule
```

DESIGN CODE OF 4-BIT RIPPLE CARRY ADDER (VERILOG)

/home/dhruv/cao_lab/vivado_proj/cao_lab_u24ev057/rtl/rca_board/rca4_verilog.v [Verilog]

```
1  `timescale 1ns / 1ps
2  //////////////////////////////////////////////////////////////////////////////////
   //
3  // Company:
4  // Engineer:
```

```
5  //
6  // Create Date: 10/01/2026 04:42:48 PM
7  // Design Name:
8  // Module Name: rca4_verilog
9  // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 //////////////////////////////////////////////////////////////////////////////////
   //
21
22
23 module rca4_verilog(
24         input [3:0] a,b,
25         input cin,
26         output [3:0] s,
27         output cout
28     );
29
30     wire [4:0] c;
31
32     assign c[0] = cin;
33     assign cout = c[4];
34
35     genvar i;
36
37     generate
38
39         for(i = 0 ; i < 4 ; i = i + 1) begin : fa
40
41             fa_verilog fa (a[i] , b[i] , c[i] , s[i] , c[i+1]);
42
43         end
44
45
46     endgenerate
47
48 endmodule
```

TESTBENCH CODE OF 4-BIT RIPPLE CARRY ADDER (VERILOG)

/home/dhruv/cao_lab/vivado_proj/cao_lab_u24ev057/tb/rca_board/rca4_tb.v [Verilog]

```
1  `timescale 1ns / 1ps
```

```
//////////////////////////////////////////////////////////////////////////////////
// Company:
// Engineer:
//
// Create Date: 10/01/2026 05:04:36 PM
// Design Name:
// Module Name: rca4_tb
// Project Name:
// Target Devices:
// Tool Versions:
// Description:
//
// Dependencies:
//
// Revision:
// Revision 0.01 - File Created
// Additional Comments:
//
//////////////////////////////////////////////////////////////////////////////////


module rca4_tb();

    reg [3:0] a , b;
    reg cin;
    wire [3:0] s;
    wire cout;
    integer i , j ,k;

    rca4_verilog rca4_test (a , b , cin , s , cout);

    initial begin

            for (i = 0 ; i < 16 ; i = i + 1) begin
                for (j = 0 ; j < 16 ; j = j + 1) begin
                    for (k = 0 ; k < 2 ; k = k + 1) begin

                        a <= i;
                        b <= j;
                        cin <= k;

                        #10;

                    end
                end
            end
 end

endmodule
```

CONSTRAINT FILE FOR BASYS 3 BOARD (XDC)

/home/dhruv/cao_lab/vivado_proj/cao_lab_u24ev057/constraints/rca/rca4.xdc [XDC]

```
## This file is a general .xdc for the Basys3 rev B board
## To use it in a project:
## - uncomment the lines corresponding to used pins
## - rename the used ports (in each line, after get_ports) according to the top level signal names in the project

## Clock signal
#set_property -dict { PACKAGE_PIN W5   IOSTANDARD LVCMOS33 } [get_ports clk]
#create_clock -add -name sys_clk_pin -period 10.00 -waveform {0 5} [get_ports clk]


## Switches
set_property -dict { PACKAGE_PIN V17   IOSTANDARD LVCMOS33 } [get_ports {a[0]}]
set_property -dict { PACKAGE_PIN V16   IOSTANDARD LVCMOS33 } [get_ports {a[1]}]
set_property -dict { PACKAGE_PIN W16   IOSTANDARD LVCMOS33 } [get_ports {a[2]}]
set_property -dict { PACKAGE_PIN W17   IOSTANDARD LVCMOS33 } [get_ports {a[3]}]
set_property -dict { PACKAGE_PIN W15   IOSTANDARD LVCMOS33 } [get_ports {b[0]}]
set_property -dict { PACKAGE_PIN V15   IOSTANDARD LVCMOS33 } [get_ports {b[1]}]
set_property -dict { PACKAGE_PIN W14   IOSTANDARD LVCMOS33 } [get_ports {b[2]}]
set_property -dict { PACKAGE_PIN W13   IOSTANDARD LVCMOS33 } [get_ports {b[3]}]
set_property -dict { PACKAGE_PIN V2    IOSTANDARD LVCMOS33 } [get_ports {cin}]
#set_property -dict { PACKAGE_PIN T3    IOSTANDARD LVCMOS33 } [get_ports {sw[9]}]
#set_property -dict { PACKAGE_PIN T2    IOSTANDARD LVCMOS33 } [get_ports {sw[10]}]
#set_property -dict { PACKAGE_PIN R3    IOSTANDARD LVCMOS33 } [get_ports {sw[11]}]
#set_property -dict { PACKAGE_PIN W2    IOSTANDARD LVCMOS33 } [get_ports {sw[12]}]
#set_property -dict { PACKAGE_PIN U1    IOSTANDARD LVCMOS33 } [get_ports {sw[13]}]
#set_property -dict { PACKAGE_PIN T1    IOSTANDARD LVCMOS33 } [get_ports {sw[14]}]
#set_property -dict { PACKAGE_PIN R2    IOSTANDARD LVCMOS33 } [get_ports {sw[15]}]


## LEDs
set_property -dict { PACKAGE_PIN U16   IOSTANDARD LVCMOS33 } [get_ports {s[0]}]
set_property -dict { PACKAGE_PIN E19   IOSTANDARD LVCMOS33 } [get_ports {s[1]}]
set_property -dict { PACKAGE_PIN U19   IOSTANDARD LVCMOS33 } [get_ports {s[2]}]
set_property -dict { PACKAGE_PIN V19   IOSTANDARD LVCMOS33 } [get_ports {s[3]}]
set_property -dict { PACKAGE_PIN W18   IOSTANDARD LVCMOS33 } [get_ports {cout}]
#set_property -dict { PACKAGE_PIN U15   IOSTANDARD LVCMOS33 } [get_ports {led[5]}]
#set_property -dict { PACKAGE_PIN U14   IOSTANDARD LVCMOS33 } [get_ports {led[6]}]
#set_property -dict { PACKAGE_PIN V14   IOSTANDARD LVCMOS33 } [get_ports {led[7]}]
```

```
#set_property -dict { PACKAGE_PIN V13   IOSTANDARD LVCMOS33 } [get_ports {led[8]}]
#set_property -dict { PACKAGE_PIN V3    IOSTANDARD LVCMOS33 } [get_ports {led[9]}]
#set_property -dict { PACKAGE_PIN W3    IOSTANDARD LVCMOS33 } [get_ports {led[10]}]
#set_property -dict { PACKAGE_PIN U3    IOSTANDARD LVCMOS33 } [get_ports {led[11]}]
#set_property -dict { PACKAGE_PIN P3    IOSTANDARD LVCMOS33 } [get_ports {led[12]}]
#set_property -dict { PACKAGE_PIN N3    IOSTANDARD LVCMOS33 } [get_ports {led[13]}]
#set_property -dict { PACKAGE_PIN P1    IOSTANDARD LVCMOS33 } [get_ports {led[14]}]
#set_property -dict { PACKAGE_PIN L1    IOSTANDARD LVCMOS33 } [get_ports {led[15]}]

```

SCREENSHOT OF CODE AND TESTBENCH

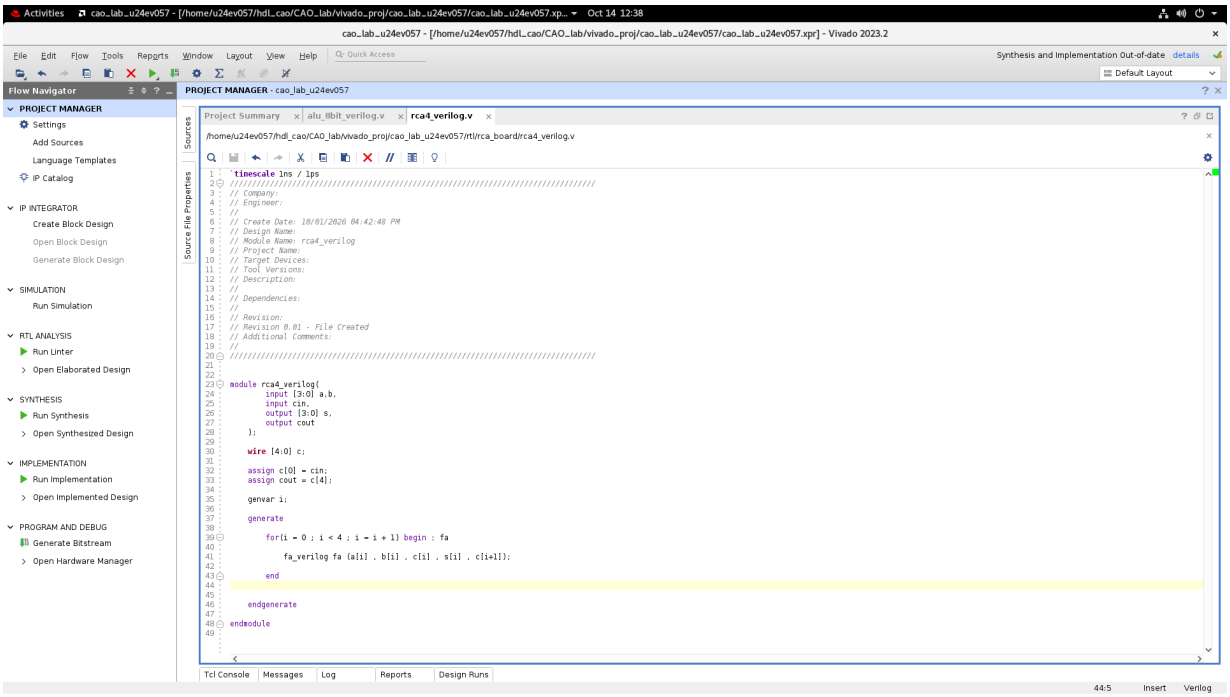


Figure 1: 4-bit RCA Verilog Design Code

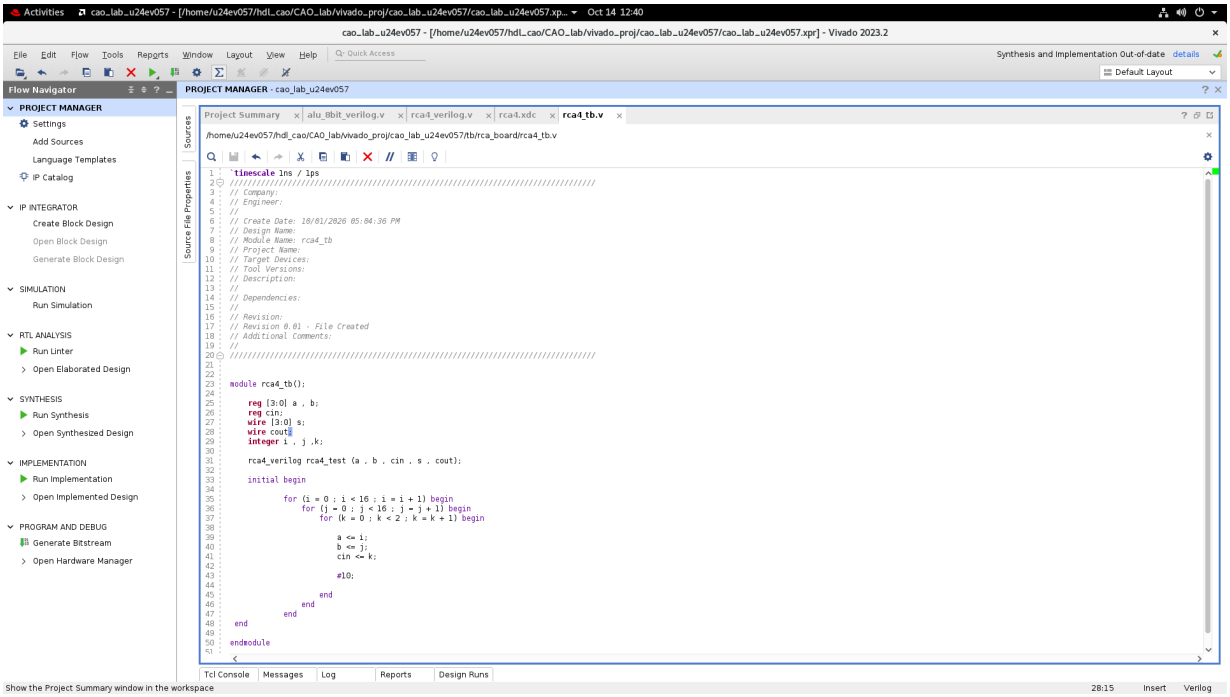


Figure 2: 4-bit RCA Testbench Code

SCREENSHOT OF CONSTRAINTS FILE

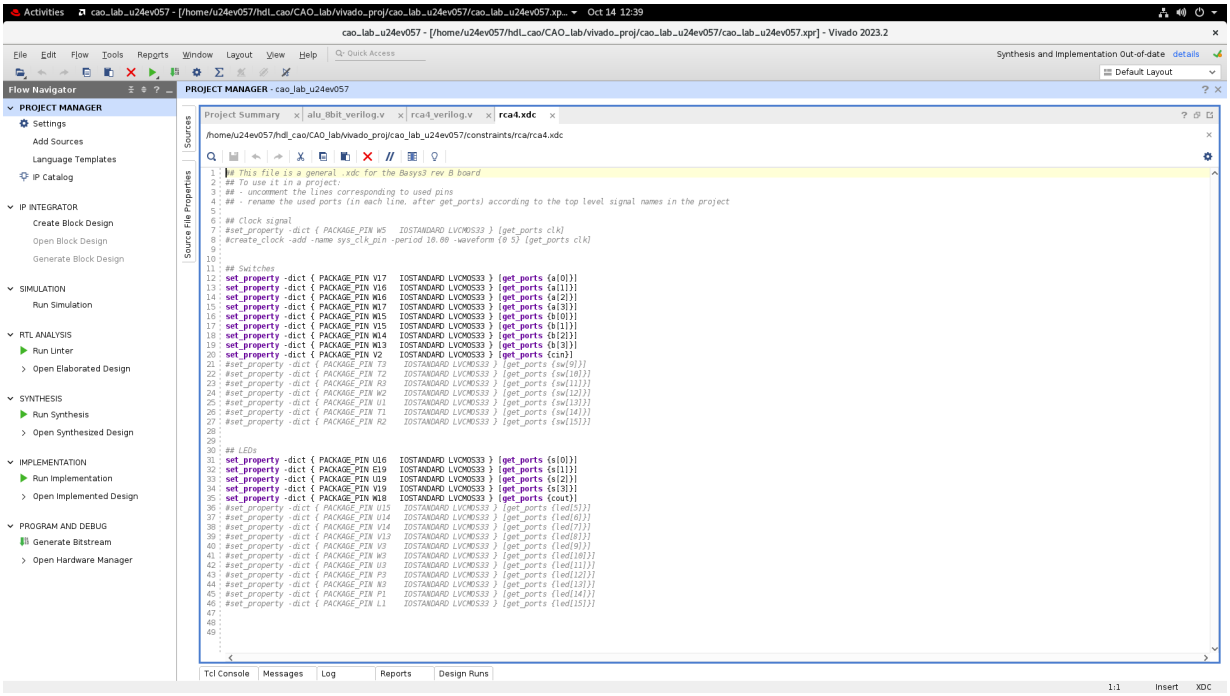


Figure 3: Basys 3 Constraints File

SIMULATION WAVEFORM

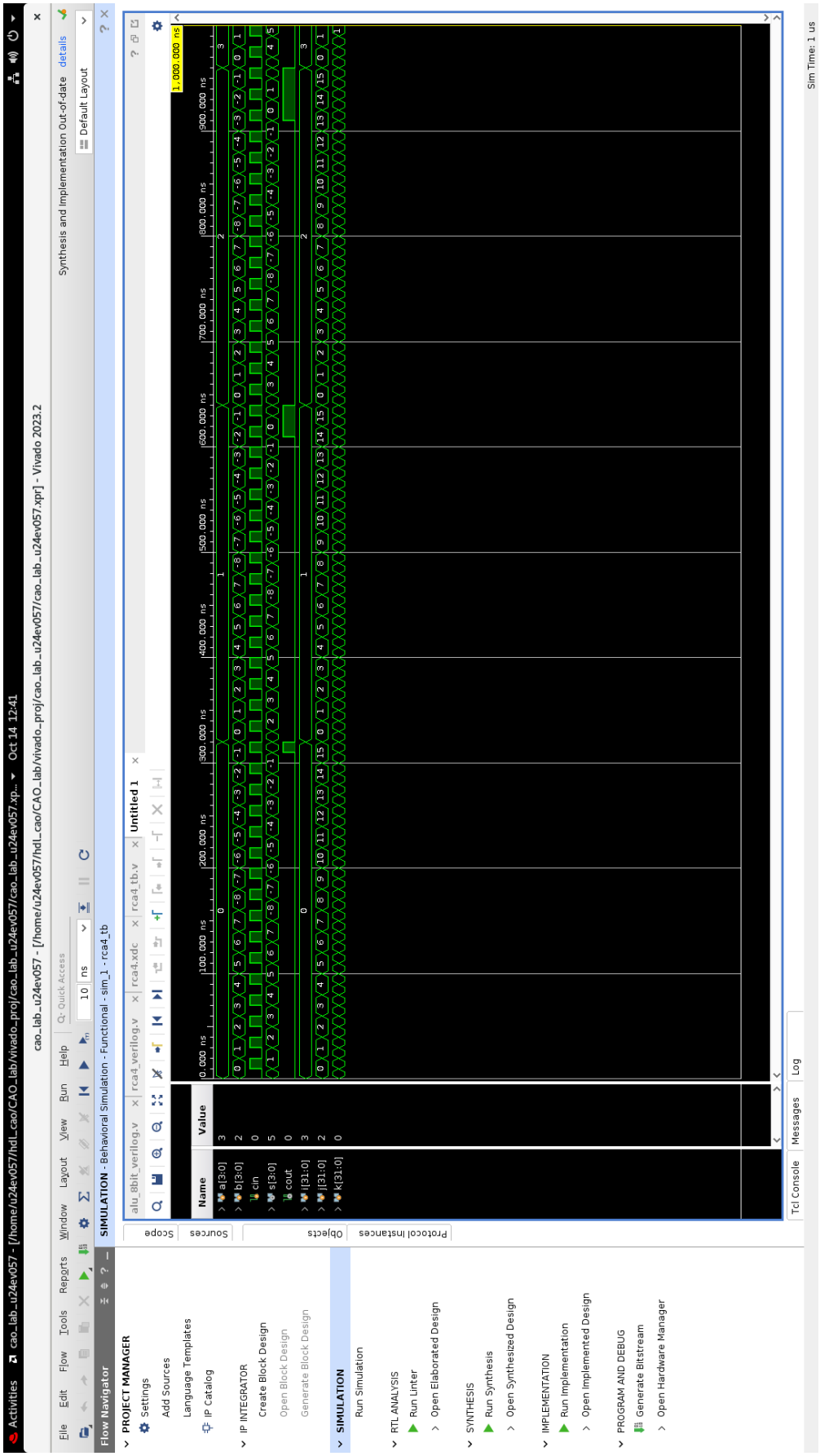


Figure 4: 4-bit RCA Simulation Waveform